How a look-alike PAIR is created

GeoPlant / geophyto_qa · image → look-alike-label diagnosis VQA

Bugwood disease table (BugWood_Diseases_enriched.csv)

host · disease · scientific name · image URL · state

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1 · MINE PAIRS

within ONE crop, take every pair of its diseases as a candidate look-alike (combinatorial)

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2 · CONFIRM THE PAIR (two-level)

WEB GATE: a credible source (extension / .edu / journal) states the two are confused → quote + URL (this decides)
CLIP: photos of the two classes are visually entangled (cross-kNN) → attached as evidence

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3 · DECISION GRAPH (per pair)

author the ONE decisive VISIBLE sign that tells member A from member B (+ lay wording, management)

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4 · VLM LABEL (per image)

for each Bugwood photo: is the deciding sign visible? close-up? which organ?

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5 · BUILD → ITEM (per sign-visible image)

render farmer↔expert dialogue: expert reads the visible sign, diagnoses, RULES OUT the look-alike

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OUTPUT ITEM

image + 6-turn dialogue + CoT (PERCEIVE→READ_SIGN→RULE_OUT→CONCLUDE) + gold {diagnosis, ruled_out, evidence_from_image}

counts

853
candidate pairs

201
→ confirmed look-alikes

151
→ pairs with items

1,453
→ dataset items

worked example

crop = cucumber → diseases {anthracnose, downy mildew, gummy stem blight, powdery mildew, ...}
candidate pair = Anthracnose × Cucurbit Downy Mildew
WEB GATE ✓ "anthracnose ... is easily confused with downy mildew" (UF/IFAS) + CLIP entangled✓
DECISION GRAPH → decisive sign: sunken lesions with salmon-pink ooze (anthracnose) vs angular spots,
fuzzy gray-purple underside (downy mildew) → each sign-visible photo → one diagnosis item